



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I - III Semester(2025-26)
Faculty Name	Ms. Alluri Swathi
Student Name	Keerthi Priya Ande
Roll Number	2520090177
Branch	CSIT
Course Outcome	5

Case Study Topic: CampusFlow – University Campus Resource Management System using Heap Sort

Work Done Summary:

Implemented Heap Sort in Java for efficient campus resource management and usage analytics. Stored campus facilities such as classrooms, laboratories, libraries, and seminar halls along with their booking counts. Applied Heap Sort to organize resources based on facility utilization and booking statistics. Implemented resource insertion, heap construction, heapify operations, and sorting functionality using a Max Heap structure. Demonstrated how campus resources can be sorted according to booking frequency to identify highly utilized and underutilized facilities. Heap Sort provides efficient performance for campus usage analytics, infrastructure planning, and resource optimization in large-scale university management systems.

Implementation Details :

1. Created a Resource class containing resource ID, resource name, and booking count.
2. Stored campus facilities such as Computer Lab, Library, Seminar Hall, Physics Lab, and Classroom A in an array.
3. Implemented Heap Sort using a Max Heap data structure.
4. Built the heap from campus resource booking data.
5. Performed heapify operations to maintain heap properties during sorting.
6. Sorted resources based on booking counts to analyze facility utilization.

7. Displayed the sorted list of campus resources after Heap Sort execution.
8. Main method inserted campus resource records and demonstrated booking-count-based sorting using Heap Sort.

Result : Screenshots/Output:

```
CampusFlow - University Campus Resource Management System
Using Heap Sort

Before Sorting:
Computer Lab - 120 bookings
Library - 95 bookings
Seminar Hall - 80 bookings
Physics Lab - 60 bookings
Classroom A - 150 bookings

Campus Resources Sorted by Booking Count:

Resource ID: 104 | Resource Name: Physics Lab | Bookings: 60
Resource ID: 103 | Resource Name: Seminar Hall | Bookings: 80
Resource ID: 102 | Resource Name: Library | Bookings: 95
Resource ID: 101 | Resource Name: Computer Lab | Bookings: 120
Resource ID: 105 | Resource Name: Classroom A | Bookings: 150

Heap Sort Completed Successfully.
```

Time Complexity:

- Heap Construction: $O(n)$
- Heapify Operation: $O(\log n)$
- Heap Sort: $O(n \log n)$
- Heap Sort efficiently manages campus resource usage data and supports infrastructure optimization and resource allocation decisions.

GitHub Link : https://github.com/keerthipriya2008/DSA2__CO5

Student Signature	Faculty Signature